

Traction Elevator Troubleshooting Guide

Rocket Elevators - Ontario Fleet Technical Reference

Overview

Traction elevators use a sheave-and-rope system driven by an AC or DC motor. Faults typically originate in the machine room (motor, controller, brakes, sheave) or in the hoistway (ropes, counterweight, safeties, guide rails). This guide addresses the most common symptoms encountered on Ontario fleet units.

Rough Ride - Diagnosis

Rough ride is the most frequent complaint. Work through causes in order:

1. Guide rail alignment: check with a plumb line - rails must be within 3 mm of plumb over any 5 m section.
2. Guide shoe wear: roller guides should have < 1 mm play; slide guides require greasing every 3 months.
3. Rope tension equalization: check with a rope-tension meter - tension between any two ropes must be within 10% of each other.
4. Sheave groove wear: measure groove depth with a groove gauge; regrind or replace sheave when groove depth exceeds 3 mm.
5. Motor/controller: check for hunting in the drive - tune the V/Hz ratio or PID gains in the drive controller.

Door Malfunctions

Doors are responsible for over 40% of all traction elevator incidents. Check the door operator belt or chain for wear and tension monthly. Door close force must not exceed 135 N (ASME A17.1 2.1.3.3). Test the door reversal device: place a 6 kg block in the door path - doors must reverse before contact force reaches 67 N. Clean door tracks quarterly; a single piece of debris can cause nuisance calls and entrapments.

Brake Adjustment

The electromagnetic brake must hold 125% of rated load when set. Inspect brake linings for wear (replace at 50% lining thickness) and for oil contamination (zero tolerance - clean with brake cleaner, identify and fix oil source). Adjust brake air gap to manufacturer specification (typically 0.3 - 0.5 mm). Brake release should be smooth; a loud clunk indicates excessive gap or worn drum surface.

Motor and Drive Faults

DC motors: check commutator for pitting, brush length (replace at 25% of new), and brush spring pressure. Clean commutator with commutator stone - never sandpaper.

AC gearless: check encoder coupling for fretting, verify encoder cable shield is grounded at one end only. Drive fault codes beginning F0xx are typically encoder-related; F3xx indicate overcurrent - check motor insulation resistance (must be > 1 M to ground at 500 V megger).

Governor and Safety System

Perform governor trip test annually per ASME A17.1 Section 2.17. Governor trip speed must be within 10% of nameplate value. After any safety application, a licensed mechanic must reset the safety wedges before the elevator is returned to service - do not use the inspection drive to re-lift a car on set safeties.